Compile time

Generated once per rule

User rule (matrix example)

```
(d: dot
  (tr: transpose A) B)
=> (out: dot A B)
```

Omitted axis fields → inference

Operation semantics

```
transpose
dot
```

TEPL compiler

Synthesize + verify

Generated attribute function

`f(m) → replacement dot attrs`

`m: shapes and captured attrs`

Runtime

Evaluate on each match

Captured attrs

$\pi = [1, 0]$

δ: left axis = 0

→ **Apply f** `f(m)` →

New dot attrs

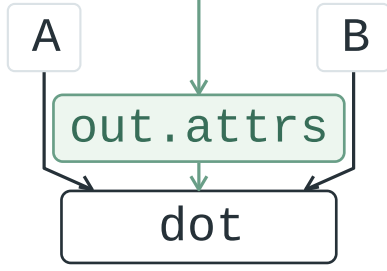
left axis = 1

`out.attrs`

Apply replacement

`(out: dot A B)`

Transpose removed



Only the left contraction axis is shown.